

SmartPay Utility Bill Generator

A simple **Core Java console-based application** that generates electricity/water utility bills using **progressive slab rates**.

This project demonstrates concepts of:

- Interfaces in Java
 - OOP principles
 - Input validation
 - Exception handling
 - Loops
 - Console-based receipt generation
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Problem Statement

A municipality wants to digitize electricity and water billing.

The application should:

- Calculate bills based on units consumed
 - Apply slab-wise pricing
 - Validate meter readings
 - Generate a digital receipt
 - Support multiple customer entries until the user exits
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Features

- 1 Progressive slab billing system
 - 2 Interface implementation (`Billable`)
 - 3 Input validation for meter readings
 - 4 Exception handling for invalid input
 - 5 Multiple customer billing support using loops
 - 6 Console-based digital receipt generation
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Technologies Used

- Java
 - OOP Concepts
 - Scanner Class
 - Exception Handling
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Project Structure

```
SmartPay/  
|— Smartpay.java  
|— README.md
```

Billing Slab Logic

Units Consumed	Rate Per Unit
0 – 100	₹1
101 – 300	₹2
Above 300	₹5

Interface Used

```
interface Billable {  
    double calculateTotal();  
}
```

The `bill_genrator` class implements the `Billable` interface and defines the bill calculation logic.

Input Validation

The program checks:

- Previous meter reading cannot be greater than current reading
- Invalid numeric input is handled using exception handling

Example:

```
Previous Reading Cannot Be Greater Than Current Reading
```

Sample Output

```
Please Enter Consumer Name ( Or Enter Exit To Quit The Application) :  
Ayush
```

```
Please Enter Current Units :  
450
```

```
Please Enter Previous Units :  
100
```

```
===== SMARTPAY RECEIPT =====  
Customer Name   : Ayush  
Units Consumed  : 350.0  
Tax Amount      : Included in slab rates  
Final Amount    : ₹750.0
```

How to Run

Step 1: Compile the Program

```
javac Smartpay.java
```

Step 2: Run the Program

```
java Smartpay
```

Concepts Covered

- Java Interfaces
- Constructors
- Encapsulation
- Exception Handling
- Loops
- Conditional Statements
- User Input Handling
- Console Formatting

Future Improvements

- GUI version using Swing/JavaFX
 - Database connectivity
 - PDF bill generation
 - GST/Tax calculation
 - Save customer history
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